

QA Report: Accounts API

Report ID: QA-PEG-003 | **Date:** 2026-04-09 | **Sprint:** Sprint 3 **Service:** Ubn-Accounts-API | **Environment:** Development (dev-api.onecluster.co)

Executive Summary

The Accounts API provides virtual account creation, balance inquiry, and statement retrieval capabilities for the Pegasus BaaS platform. In sandbox mode, the service returns deterministic mock responses with a fixed balance of 500,000 NGN to enable partner integration testing. This report covers QA testing of all Accounts API endpoints on the development environment.

Scope

- POST /api/v1/accounts — Create virtual account
- GET /api/v1/accounts/{accountNumber} — Balance inquiry
- GET /api/v1/accounts/{accountNumber}/statement — Account statement with date filters
- Sandbox mock response behavior (fixed balances, test account names)
- HMAC-SHA256 signature validation
- Error handling, validation, and edge cases

Changes This Sprint (Mar 31 — Apr 9)

- 18 commits to Ubn-Accounts-API
- Full platform sandbox APIs added across all account endpoints
- CI/CD pipeline configuration and deployment automation
- Sandbox mock data seeding for deterministic test scenarios
- Response envelope standardization
- Logging and audit trail enhancements
- Health check endpoint added

Test Results

Virtual Account Creation

Test ID	Test Case	Expected	Actual	Status	Notes
QA-ACC-001	Create virtual account with valid payload	201 with accountNumber	201 with accountNumber, accountName:"SANDBOX TEST ACCOUNT", currency:"NGN"	PASS	Sandbox returns SB-prefixed account numbers
QA-ACC-002	Create virtual account with missing fields	422 Validation error	422 with field-level errors (accountName, customerEmail required)	PASS	Proper RFC 7807 validation
QA-ACC-003	Create virtual account without HMAC	401 HMAC_REQUIRED	401 with error:"HMAC signature is required"	PASS	Security gate enforced
QA-ACC-004	Create virtual account with invalid currency	422 UNSUPPORTED_CURRENCY	422 with error:"Only NGN is supported"	PASS	Currency whitelist active

QA-ACC-005	Create duplicate virtual account (same reference)	409 Conflict or idempotent 200	200 with existing account details returned	PASS	Idempotency on reference key
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Balance Inquiry

Test ID	Test Case	Expected	Actual	Status	Notes
QA-ACC-006	Get balance for valid account	200 with balance details	200 with availableBalance:500000, ledgerBalance:500000, currency:"NGN"	PASS	Sandbox returns fixed 500,000 NGN balance
QA-ACC-007	Get balance for non-existent account	404 ACCOUNT_NOT_FOUND	404 with error:"Account not found"	PASS	Proper error for invalid accounts
QA-ACC-008	Get balance without authentication	401 Unauthorized	401 with error:"API key is required"	PASS	Auth enforcement verified
QA-ACC-009	Get balance with invalid HMAC	401 HMAC_INVALID	401 with error:"HMAC signature validation failed"	PASS	Tampered requests rejected
QA-ACC-010	Balance response field types	availableBalance as number	availableBalance:500000 (number, not string)	PASS	Correct JSON typing

Account Statement

Test ID	Test Case	Expected	Actual	Status	Notes
QA-ACC-011	Get statement for valid account	200 with transactions array	200 with transactions:[], pagination metadata	PASS	Returns empty array for new sandbox accounts
QA-ACC-012	Get statement with date range	200 with filtered transactions	200 with transactions filtered by startDate/endDate	PASS	Date filtering works correctly
QA-ACC-013	Get statement with invalid date format	422 INVALID_DATE_FORMAT	422 with error:"Date must be in YYYY-MM-DD format"	PASS	Date validation enforced
QA-ACC-014	Get statement for non-existent account	404 ACCOUNT_NOT_FOUND	404 with error:"Account not found"	PASS	Consistent error handling
QA-ACC-015	Get statement with pagination	200 with page metadata	200 with page, pageSize, totalCount, totalPages	PASS	Pagination metadata present

QA-ACC-016	Get statement with large page size	200 capped at max	200 with pageSize capped at 100	PASS	Server-side cap prevents abuse
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Sandbox Mock Data

Test ID	Test Case	Expected	Actual	Status	Notes
QA-ACC-017	Sandbox account name	"SANDBOX TEST ACCOUNT"	"SANDBOX TEST ACCOUNT"	PASS	Clearly identifies sandbox data
QA-ACC-018	Sandbox balance consistency	Same balance on repeated calls	500000 NGN on every call	PASS	Deterministic sandbox behavior
QA-ACC-019	Sandbox account number prefix	SB- prefix	Account numbers start with SB-	PASS	Easy to distinguish sandbox vs live

Issues Found

ID	Severity	Issue	Status
PEG-391	Low	Account statement returns empty transactions array for all sandbox accounts — would be more useful with seeded mock transactions for partner testing	OPEN
PEG-392	Low	Balance response does not include lastUpdated timestamp — partners cannot determine data freshness	OPEN

Security Verification

Check	Expected	Actual	Status
HMAC-SHA256 on all endpoints	Required	Enforced on POST and GET endpoints	PASS
API key authentication	Required	x-api-key header validated	PASS
Rate limiting	Per-partner limits	Rate limiting headers present (X-RateLimit-Remaining)	PASS
PII in logs	Masked	Account numbers and customer names masked in application logs	PASS

Recommendations

1. Seed sandbox accounts with 5-10 mock transactions to enable statement testing without requiring real inbound transfers
2. Add `lastUpdated` timestamp to balance response for data freshness indication
3. Consider adding bulk account creation endpoint for partners managing high volumes
4. Document sandbox account number format (SB- prefix) in partner integration guide